



Applied Continuous Learning in the Lifecycle of AI-Enabled Software Systems: A Survey

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Abstract

In recent years, AI-enabled software systems have transformed numerous industries. However, maintaining the accuracy and relevance of these systems over time remains a challenge, as data distributions (data drift), user needs, and environmental conditions evolve. Continuous learning (CL) addresses this issue by allowing AI models to learn and adapt incrementally without forgetting previously acquired knowledge. This paper presents a systematic overview of continuous learning techniques in AI-enabled software systems, focusing on current methodologies, challenges, and future research directions.

1. Introduction

As artificial intelligence (AI) systems become integral to critical domains such as healthcare, finance, and manufacturing, the demand for adaptive systems capable of maintaining performance over time has grown significantly. Traditional AI models are typically trained on static datasets, which can limit their effectiveness when encountering new or unseen data that deviates from the original training distribution. Continuous learning (CL), also referred to as lifelong learning, addresses this limitation by enabling AI systems to incrementally update their knowledge while retaining previously learned information [1]. This capability allows AI systems to dynamically adapt to evolving environments, thereby improving their long-term performance.

Despite its advantages, CL presents significant challenges, the foremost being **catastrophic forgetting**, where learning new information disrupts previously acquired knowledge [2]. Other challenges include ensuring computational efficiency, maintaining scalability, addressing data privacy concerns, and guaranteeing model reliability over time. Developing effective CL strategies is crucial for advancing AI-enabled software systems that operate in dynamic and rapidly changing environments.

2. Background

2.1 Continuous Learning methodologies in AI

Current research on continuous learning has explored various methodologies including Meta-Learning, Incremental Learning, Dynamic Architecture Search, Self-Supervised Learning, and Reinforcement Learning.

2.1.1 Meta-Learning

Meta-learning, often referred to as "learning to learn," focuses on training models to generalize learning strategies across tasks rather than optimizing for individual tasks. This approach enhances a model's ability to adapt to novel tasks with minimal data, as exemplified by the Model-Agnostic Meta-Learning (MAML) framework [3]. By emphasizing adaptability, meta-learning is particularly advantageous for task-specific applications where data is scarce. However, it has notable limitations, including high computational demands and a dependency on well-curated task datasets. Despite these challenges, the method remains a powerful tool for scenarios requiring efficient task adaptation.

2.1.2 Incremental Learning

Incremental learning is a paradigm designed to enable models to adapt to new data while retaining previously learned knowledge, effectively mitigating the issue of catastrophic forgetting. Techniques such as Elastic Weight Consolidation (EWC) achieve this by employing regularization to preserve critical parameters essential for prior tasks [4]. This approach strikes a balance between stability and plasticity, making it well-suited for dynamic environments where data evolves over time. However, incremental learning can face significant performance challenges, particularly when confronted with large shifts in data distribution.

2.1.3 Dynamic Architecture Search

Dynamic Architecture Search (DAS) leverages Neural Architecture Search (NAS) techniques to dynamically adapt network structures, optimizing performance for tasks that evolve over time. This adaptability makes DAS particularly efficient for multi-task learning and scenarios requiring automated adaptation to changing requirements [5]. Despite its strengths, DAS is computationally intensive, demanding significant resources, and presents challenges in interpretability, complicating the understanding of its learned architectures.

2.1.4 Self-Supervised Learning

Self-supervised learning is a paradigm that derives supervisory signals from the data itself, facilitating the use of extensive unlabeled datasets for pre-training. This methodology is particularly advantageous for continuous learning, where datasets evolve over time [6]. By significantly reducing the reliance on human-labeled data, self-supervised learning enables large-scale learning across diverse applications. However, its performance can be constrained when applied to domain-specific or non-representative datasets, limiting its generalizability in such contexts.

2.1.5 Reinforcement Learning

Reinforcement Learning (RL) focuses on enabling agents to make sequential decisions that maximize cumulative rewards over time. In the context of continuous learning, RL systems utilize techniques such as policy distillation and transfer learning to enhance adaptability across evolving scenarios [7]. RL is particularly well-suited for dynamic environments involving sequential decision-making. However, it faces notable limitations, including sample inefficiency and high computational demands, which can hinder scalability in real-world applications.

3. Methodology

3.1 Research Goals Formulation

The primary objective of this survey is to provide a comprehensive and systematic review of diverse approaches within Continual Learning (CL), emphasizing their relevance and applicability to AI-enabled software systems. The study examines **RQ1. What are the respective strengths and identifies potential limitations of each approach.**

Additionally, the research seeks to address two critical dimensions:

RQ2. Integration Challenges: Identify and analyze the potential challenges in integrating CL methodologies into AI-enabled software systems, particularly in dynamic and evolving

environments. This includes examining constraints related to computational resources, system scalability, data privacy, and real-time adaptability within the software lifecycle.

RQ3. Emerging Trends and Future Directions: Explore the emerging trends and future directions of applied CL in the lifecycle of software systems. This involves investigating how CL can be leveraged to enhance software capabilities, such as enabling adaptive maintenance, improving fault tolerance, and optimizing resource utilization, while also identifying promising avenues for further research and innovation.

3.2 Search Strategy

To ensure a comprehensive review, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework was adopted to identify, screen, and select relevant literature. The survey utilized the following scholarly databases and platforms to gather high-quality and relevant papers: **ACM Computing Surveys**, **IEEE Xplore** and **Google Scholar**. These sources were chosen due to their extensive repositories of computing surveys, conference papers, and authoritative recent research in computer science and AI.

3.3 Inclusion and Exclusion Criteria

The survey aims to give readers a good understanding of current CL methodologies and identify potential limitations. Further emphasizing their relevance and applicability to AI-enabled software systems life cycle and analyze the potential challenges in integrating and future direction. Following are the added inclusion criteria:

- Papers published in peer-reviewed journals or recognized conferences.
- Studies focusing on Continual Learning methodologies.
- Papers discussing integration challenges or emerging trends in AI-enabled software systems.
- Publications within the last 10 years to ensure relevance to current technological advancements.

However, the surveyed query terms return many irrelevant works to this review. Some papers may contain only CL and cover completely unrelated topics. The exclusion criteria are listed below:

- Papers that do not explicitly focus on Continual Learning.
- Papers that are only focused on introducing optimizing technique to enhance the performance of existing continual learning (CL) methodologies,
- Studies with insufficient discussion on applications of CL.

4. Related study

4.1 Applied CL in Recommendation Systems

Recommendation systems in platforms like Netflix, Amazon, and YouTube have incorporated continuous learning to provide users with relevant content as preferences evolve over time. For instance, Amazon's recommendation engine uses customer behavior data to adjust product

suggestions on the fly. A recent review [8] introduced a deep reinforcement learning based recommender system (RLRS) formulated user recommendation as a **Markov decision process (MDP)**. The survey proposed an RLRS framework with four components, i.e., state representation, policy optimization, reward formulation, and environment building, and survey RLRS algorithms accordingly. **State Representation:** restructuring the user interactions and preferences in a format suitable for reinforcement learning algorithms. **Policy Optimization:** Algorithm such as Deep Q-Networks (DQN), Policy Gradient, or Actor-Critic methods to derive optimal recommendation strategies that maximize cumulative rewards over user interactions. **Reward Formulation:** Rewards are designed to capture user satisfaction, such as clicks, time spent, or purchases. Complex reward formation considers longer-term engagement metrics e.g. session completion rate, retention rate etc. **Environment Building:** simulates user interactions in a controlled environment for training RL agents. This includes building user models and simulating diverse interaction patterns.

The survey also highlights significant challenges in integrating RLRS into real-world recommendation systems. **Data Sparsity and Scalability:** Real-world recommendation environments face high-dimensional user-item spaces, resulting in sparse reward signals. Addressing this requires efficient data sampling techniques and robust exploration strategies. **Cold Start Problem:** RLRS frameworks encounter difficulties with new users or items due to the lack of interaction history, impeding policy training and early performance. **Computational Overhead:** Training and deploying RL-based models, particularly deep RL, necessitate significant computational resources and infrastructure to handle dynamic and large-scale user interactions.

4.2 Applied CL in manufacturing environment monitoring Systems

CL in manufacturing environment monitoring systems plays a critical role in enhancing the adaptability and efficiency of industrial artificial intelligence (AI) applications. In the context of Industry 4.0, CL enables AI systems to incrementally update and refine their models by integrating new data generated from dynamic manufacturing environments [9]. This ensures that AI solutions remain robust and effective despite the evolving complexities of industrial processes. The research emphasizes that CL addresses challenges such as data variability, equipment degradation, and changes in operational conditions, which are common in manufacturing settings. By leveraging CL, manufacturing monitoring systems can achieve real-time adaptability and maintain high performance, contributing to improved decision-making and process optimization within smart factories.

CL offers significant advantages but also presents several challenges. One key challenge is characterized by heterogeneous data sources and dynamic operating conditions of manufacturing settings [9]. The variability in data quality, structure, and frequency necessitates robust preprocessing and adaptable learning mechanisms to ensure the system's effectiveness. Additionally, managing computational costs and resource constraints becomes critical, as CL systems require ongoing updates, which can strain the infrastructure of industrial environments. The research also highlights the difficulty in balancing learning performance with data privacy and security, as manufacturing environments often deal with sensitive and proprietary information. Addressing these challenges is essential to fully realize the potential of CL in creating resilient and adaptive monitoring systems in Industry 4.0.

4.3 Applied CL in Healthcare and Diagnostic Systems

In the healthcare sector, AI-driven diagnostic tools and medical imaging software use continuous learning to improve diagnostic accuracy over time. For example, COVID-19 detection models trained on initial datasets had to be updated to incorporate new variants and emerging symptoms. A study [10] on continuous learning in medical imaging that presents an extensive survey of existing research and future research directions. By incrementally learning from new imaging data, these models remain robust and reliable over time, enhancing patient outcomes and adapting to the rapid changes in disease progression.

The research outlines several challenges faced in applying continual learning (CL) to medical imaging system includes, **Medical Data Drift:** The dynamic and evolving nature of medical data, such as advancements in imaging technologies, variations in patient demographics, and changes in disease characteristics, complicates model adaptation and robustness. **Heterogeneous Data Sources:** Differences in imaging modalities, acquisition techniques, and pre-processing methods across institutions and devices create inconsistencies in data quality and distribution, making generalization difficult. **High Computational and Storage Requirements:** Retraining models from scratch to incorporate new data can be computationally expensive and impractical for deployment in real-world healthcare settings. **Privacy Concerns:** Storing and using historical patient data for model updates raises significant privacy and regulatory issues, particularly in healthcare.

5. Critical Evaluation

Continuous Learning (CL) methodologies demonstrate significant strengths but come with notable limitations. The table below compares key CL methodologies, summarizing their strengths and limitations:

Table 1. Comparison of Continuous Learning (CL) methodologies Strengths and Limitations

Methodology	Strengths	Limitations
Meta-Learning	Efficient task adaptation, minimal data requirement	Computational cost, task dataset dependence
Incremental Learning	Stability-plasticity balance	Degrades with divergent data distributions
Reinforcement Learning	Sequential decision-making, adaptable policies	Sample inefficiency, computational demands
Dynamic Architecture	Efficient multi-task learning	Resource-intensive, interpretability issues
Self-Supervised Learning	Unlabeled data utilization	Limited performance in specific datasets

Applications of CL in real-world software systems highlight domain-specific challenges. In recommendation systems, CL ensures user preferences evolve dynamically, as demonstrated by reinforcement learning-based recommender systems [8]. However, challenges such as data sparsity, cold-start problems, and computational overhead hinder adoption. Manufacturing monitoring systems employ CL for real-time adaptability in Industry 4.0 environments, addressing

equipment degradation and operational variability. Despite its advantages, challenges like heterogeneous data sources, dynamic conditions, and resource constraints limit scalability and performance [9]. In healthcare, CL enables medical imaging systems to adapt to dynamic data, such as evolving disease characteristics and imaging technologies. Challenges include medical data drift, data heterogeneity, high computational demands, and privacy concerns [10].

To overcome these challenges, future research must prioritize optimizing computational efficiency and scalability of CL algorithms for large-scale applications. Developing robust frameworks for handling heterogeneous data sources and addressing data drift is critical. Privacy-preserving techniques, such as federated or encrypted learning mechanisms, must be integrated to ensure regulatory compliance in sensitive domains like healthcare. Interdisciplinary approaches combining advancements in reinforcement learning, dynamic architectures, and meta-learning could address cold-start and reward-signal sparsity issues in recommendation systems. In manufacturing, leveraging advanced preprocessing techniques and adaptive mechanisms can enhance performance in dynamic environments. For healthcare, establishing standardized data formats and federated learning frameworks can improve generalization and privacy. By addressing these areas, CL systems can achieve resilient and scalable performance across diverse applications.

Declaration:

I declare that I have personally prepared this assignment. The work is my own, carried out personally by me unless otherwise stated and has not been generated using Artificial Intelligence tools unless specified as a clearly stated approved component of the assessment brief. All sources of information, including quotations, are acknowledged by means of the appropriate citations and references. I declare that this work has not gained credit previously for another module at this or another University.

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I understand that by submitting this assessment, I declare myself fit to be able to undertake the assessment and accept the outcome of the assessment as valid.

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